

Course Plan and Policies
EM-613: Mathematical Models for Data Science
Module-I, Jun-Sep 2024
Indian Institute of Technology Gandhinagar

Instructors: 1. Pratyush Dayal Office: Academic Block 5, Room 404B Email: pdayal@iitgn.ac.in 2. Udit Bhatia Office: Academic Block 6, Room 334 Email: bhatia.u@iitgn.ac.in	Teaching Assistant Sarath Dubey Email: dubey_sarth@iitgn.ac.in
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Class Location and Meeting Times

Sessions:

Thursday, Timings 8:30 PM to 10:00 PM

Saturday: Timings 11:00AM to 12:30 PM

Mode of Engagement:

Online on Zoom. Sign-in using your IITGN email ID to connect [[link](#)]

Reference Texts

- Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong. *Mathematics for Machine Learning*. Cambridge University Press, 2020.
- Van Der Aalst, Wil, and Wil van der Aalst. *Data science in action*. Springer Berlin Heidelberg, 2016.

Course Website

URL: <https://sites.google.com/iitgn.ac.in/dsdm-em-613/home>

The course will be fully managed through the website and all the major announcements, lecture notes, scheduled activities will be uploaded to the website only. Students are advised to check the website regularly for any updates regarding the course.

Course Objectives

The aim of the course is to enable students to develop foundational understanding and skills towards data exploration, preprocessing, and statistical analysis.

Expected Outcomes

At the end of this course the students should be able to understand

- Data manipulation, including importing, cleaning, and preprocessing data using relevant programming languages and libraries.
- Various statistical methods for data analysis and hypothesis testing.
- Basics of machine learning and its application in data analysis.
- Model evaluation techniques and appropriate metrics for assessing the performance of data science models.
- Gain experience in applying acquired skills to solve real-world problems, thereby enhancing their capacity to communicate data analysis results clearly and effectively.

Learning Objectives

The course is designed to achieve following learning objectives:

- Learn how to clean and prepare data effectively for analysis.
- Understand statistical methods to draw meaningful insights from data.
- Creating clear and engaging data visualizations.
- Gain practical skills in optimizing machine learning models.

Honour Code and Plagiarism

Any form of copying in assignments/quizzes/examinations will be considered against the honor code of the Institute and will attract severe penalties. In order to know more about plagiarism and its different forms, please go [here](#) and [here](#). Please keep the following points in mind:

- The course encourages discussions between the students and the sharing of ideas. In doing so, however, it should be borne in mind that the student MUST submit their own work.
- All the assignments are checked for plagiarism using online tools.
- The students who are found copying will be awarded “F” grade.
- The students who break the honour code of the Institute will be reported to SSAC for further action. It has been generally observed that students, who break the honour code of the Institute, are typically suspended by SSAC for a minimum period of one module.

Grading (To be revisited upon starting of the class)

The final grade shall be awarded based on the total score of the student that will be calculated using the following weights.

- Quizzes: 25%
- Assignments/Tutorials: 25%
- Examination-I: 20%
- Examination-II: 20%
- Attendance/Class participation: 10%

It should be noted that the above distribution is tentative and can be modified during the module by the instructor based on student feedback. Changes, if any, shall be adequately discussed in the class and a consensus between the instructor and the students will be obtained, before implementation.

Quizzes

There will be about 4-5 quizzes, which might be taken in the beginning, middle or end of the lecture. Typically, these quizzes will have conceptual questions that will require short answers; the form of the questions shall be: one-line, multiple-choice or fill-in the blanks. The questions in the quiz might either be from the content already covered in the class or the content that is expected to be covered in the immediate next few classes. Each quiz will carry equal weight irrespective of its maximum marks.

Assignments/Tutorials

There will be about 3-4 assignments in the course. The questions in the assignment will be based on application of the concepts learned in the course for arriving at solutions to problems. Most of the assignments will require computer programming. Students are encouraged to discuss the problems with peers, TAs and instructors before submission. The following points must be noted:

- Programming assignments must use MATLAB as a programming language.
- Programs that do not run will not be evaluated.
- All the assignments will have equal weights, irrespective of its maximum marks.
- Some of the questions in the assignments might carry extra credits. These marks will be scaled and added directly to the total score that the student receives in the course.

Submission of Assignments

- All the assignments must be submitted using the Online Forms link given in the assignment.
- Assignments submitted via any other means will not be accepted.

Examinations

The assessments for Examinations shall be done through an open book exam. The students shall be given a question paper and they will be required to upload their answers in one day. After the students upload their answer books, there shall be a viva based on the answers they have submitted. The grades shall be awarded to each student, which shall be converted into marks using the adjoining table:

Letter Grade	A+	A	A-	B	B-	C	C-	D	F
Grade Point	11	10	9	8	7	6	5	4	0

Attendance and Class Participation

It is expected from the students to ask questions and participate in the discussions. The students who typically remain passive in the class will be awarded zero marks. The marks for the attendance will be calculated based on the ratio of number of classes attended and total number of lectures.

EM-613: Mathematical Models for Data Science**Instructor: Pratyush Dayal/Udit Bhatia****Sessions: Thursday: 8:30 PM – 10:00 PM, Saturday at 11:00 AM – 12:30 PM**

Week No.	Start Date	End Date	L/T No.	L/T Date	Day	Contents
Instructor: Pratyush Dayal						
W1	27-May-24	02-Jun-24	L-01	01-Jun-24	Sat	Course Overview and Policies
W2	03-Jun-24	09-Jun-24	L-02	06-Jun-24	Thu	MATLAB -- Basic Usage, Control Statements, Plotting
			T-01	07-Jun-24	Fri	Tutorial on MATLAB
			L-03	08-Jun-24	Sat	Vectors and Matrices: Definitions, properties, operations, applications in data science.
W3	10-Jun-24	16-Jun-24	L-04	13-Jun-24	Thu	Eigenvalues and Eigenvectors: Spectral decomposition, principal component analysis.
			L-05	14-Jun-24	Fri	Solving Linear Systems: Matrix factorizations, iterative methods, conditioning.
			T-02	15-Jun-24	Sat	Tutorial on Eigenvalues, Eigenvectors, Matrix factorizations
W4	17-Jun-24	23-Jun-24	L-06	20-Jun-24	Thu	Ordinary Least Squares (OLS): Assumptions, estimation, interpretations.
			L-07	22-Jun-24	Sat	Matrix Decompositions: Cholesky, QR Decomposition, Spectral, Singular Value: Algorithm, properties, applications in linear regression.
W5	24-Jun-24	30-Jun-24	T-03	27-Jun-24	Thu	Matrix Decomposition Applications: Linear regression (Cholesky), least squares fitting (QR), data analysis (Spectral), image processing (SVD)
			L-08	29-Jun-24	Sat	Convex Optimization: Definitions, properties, local and global optima.
W6	01-Jul-24	07-Jul-24	L-09	04-Jul-24	Thu	Gradient Descent Methods: Batch, stochastic, mini-batch versions, momentum, adaptive learning rates.
			T-04	06-Jul-24	Sat	Optimization Applications in Machine Learning: Training neural networks, tuning hyperparameters.
W7	08-Jul-24	14-Jul-24	ONLY Evaluations: No Lectures and/or Tutorials			
Instructor: Udit Bhatia						
W8	15-Jul-24	21-Jul-24	L-10	18-Jul-24	Thu	Python: Basic usage for understanding data
			L-11	20-Jul-24	Sat	Data Pre-processing and Feature Engineering
W9	22-Jul-24	28-Jul-24	T-05	25-Jul-24	Thu	Tutorial on Exploratory Data Analysis using Python
			L-12	27-Jul-24	Sat	Data Visualization: Storytelling with data
W10	29-Jul-24	04-Aug-24	L-13	01-Aug-24	Thu	What to expect from Mathematical Models?
			L-14	03-Aug-24	Sat	Model Evaluation: Understanding the performance metrics
W11	05-Aug-24	11-Aug-24	T-06	08-Aug-24	Thu	Tutorial: Model Evaluation
			L-15	10-Aug-24	Sat	Model Validation: Approaches for cross-validation
W12	12-Aug-24	18-Aug-24	T-07	15-Aug-24	Thu	Tutorial: Model Validation
			L-16	17-Aug-24	Sat	Model Tuning and Selection
W13	19-Aug-24	25-Aug-24	T-08	22-Aug-24	Thu	Tutorial: Model Tuning
			T-09	24-Aug-24	Sat	Tutorial: Model Selection
W14	26-Aug-24	01-Sep-24	ONLY Evaluations: No Lectures and/or Tutorials			